

12 (**long**) A two-degree-of-freedom system is shown in Fig. 9 comprising two masses m and $2m$ described by displacements y_1 and y_2 respectively. The masses are linked by springs k and $2k$ as shown. A force f is applied to mass m .

(a) Find a pair of differential equations describing the motion of the system and express this in matrix form. Identify the mass and stiffness matrices. [8]

(b) For the case $f(t) = F \cos \omega t$ find expressions for Y_1 and Y_2 as functions of frequency where $y_1(t) = Y_1 \cos \omega t$ and $y_2(t) = Y_2 \cos \omega t$. [8]

(c) Show that the two resonance frequencies for the system are at $\omega = \sqrt{2 \pm \sqrt{3}} \sqrt{\frac{k}{m}}$. [6]

(d) Sketch a graph of Y_1 and Y_2 as functions of frequency, noting carefully the two resonance frequencies and the behaviour at $\omega = \sqrt{\frac{k}{m}}$. [8]

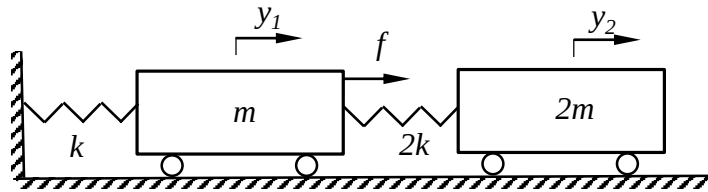


Fig. 9

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